

Key features of parabolas



1 Consider the general quadratic equation $y = ax^2 + bx + c$, $a \neq 0$.

a If $a < 0$, in what direction will the parabola open?

b If $a > 0$, in what direction will the parabola open?

2 Does the parabola represented by the equation $y = x^2 - 8x + 9$ open upward or downward?

3 Does the graph of $y = x^2 + 6$ have any x -intercepts? Explain your answer.

4 State whether the following parabolas have x -intercepts:

a $y = (x - 7)^2 + 4$

b $y = -(x - 7)^2 + 4$

c $y = -(x - 7)^2 - 4$

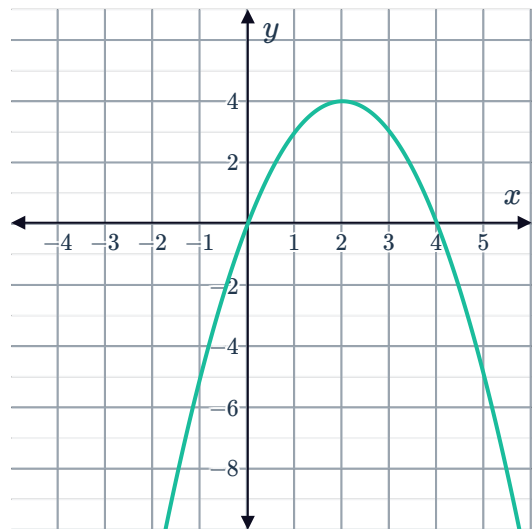
d $y = (x - 7)^2 - 4$

5 Consider the given graph:

a State the x -intercepts.

b State the y -intercept.

c State the maximum value.



6 Consider the given graph:

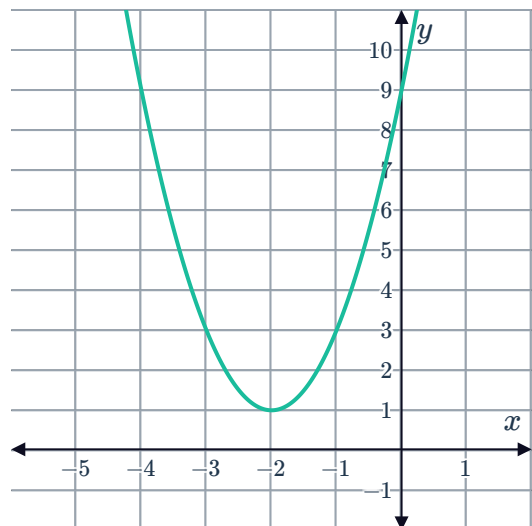
a Is the curve concave up or concave down?

b State the y -intercept of the graph.

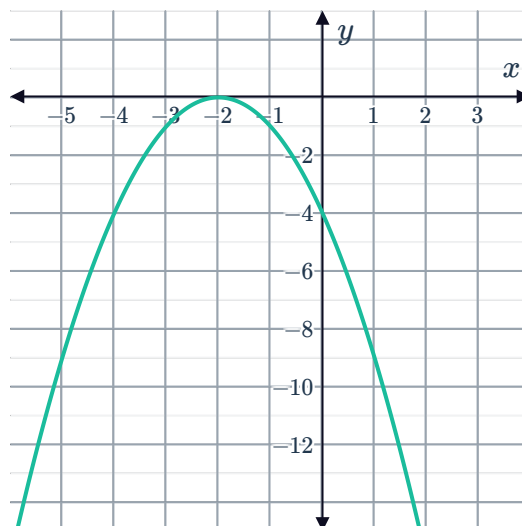
c What is the minimum value?

d At which value of x does the minimum value occur?

e Determine the interval of x for which the graph is decreasing.



7 Consider the graph of the parabola:



- a State the coordinates of the x -intercept.
- b State the coordinates of the vertex.
- c State whether the following statements are true about the vertex:
 - i The vertex is the minimum value of the graph.
 - ii The vertex occurs at the x -intercept.
 - iii The vertex lies on the axis of symmetry.
 - iv The vertex is the maximum value of the graph.

8 Suppose that a particular parabola is concave down, and its vertex is located in quadrant 2.

- a How many x -intercepts will the parabola have?
- b How many y -intercepts will the parabola have?

9 Suppose that a particular parabola has two x -intercepts, and its vertex is located in quadrant 4. Will such a parabola be concave up or concave down?

10 Consider the quadratic function defined in the table on the right:

- a What are the coordinates of the vertex?
- b What is the minimum value of the function?

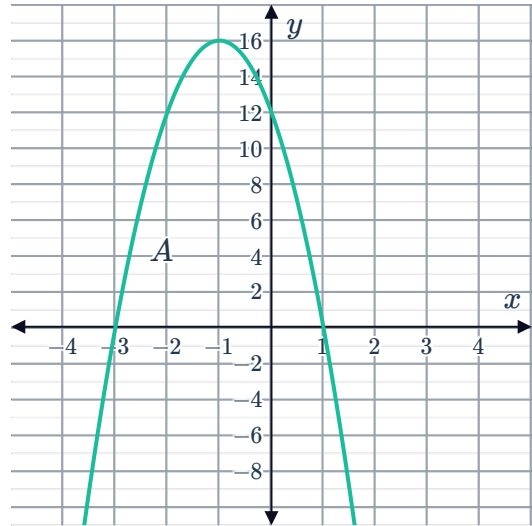
x	y
-7	11
-6	6
-5	3
-4	2
-3	3
-2	6
-1	11

11 A vertical parabola has an x -intercept at $(-1, 0)$ and a vertex at $(1, -6)$. Find the other x -intercept.

12 State whether the following can be found, without any calculation, from the equation of the form $y = (x - h)^2 + k$ but not from the equation of the form $y = x^2 + bx + c$:

- a x -intercepts
- b y -intercept
- c vertex

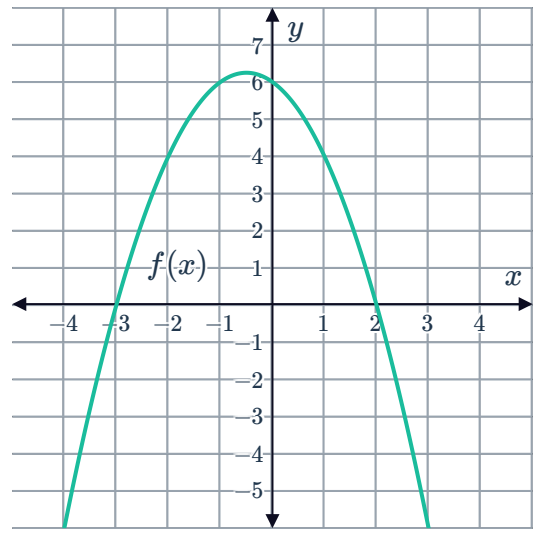
- 13 Quadratic function A is represented graphically as shown. Quadratic function B , which is concave down, shares the same x -intercepts as quadratic function A , but has a y -intercept closer to the origin. Which of the functions has a greater maximum value?



- 14 State the axis of symmetry of the parabola $y = k(x - 7)(x + 7)$ for any value of k .
- 15 Consider the equation $y = 25 - (x + 2)^2$. What is the maximum value of y ?
- 16 Consider the function $y = (14 - x)(x - 6)$.
- State the zeros of the function.
 - Find the axis of symmetry.
 - Is the graph of the function concave up or concave down?
 - Determine the maximum y -value of the function.
- 17 Consider the parabola of the form $y = ax^2 + bx + c$, where $a \neq 0$. Complete the following statement:
The x -coordinate of the vertex of the parabola occurs at $x = \mathbb{I}$. The y -coordinate of the vertex is found by substituting $x = \mathbb{I}$ into the parabola's equation and evaluating the function at this value of x .
- 18 Find the x -coordinate of the vertex of the parabola represented by $P(x) = px^2 - \frac{1}{2}px - q$.



- 19 Consider the graph of the function $f(x) = -x^2 - x + 6$:
Using the graph, write down the solutions to the equation $-x^2 - x + 6 = 0$.



20 True or false:

- a The quadratic formula can be used to find the y -intercept.
- b If the parabola has only one x -intercept, then the x -intercept is also the vertex.

- 21 Consider the parabola whose equation is $y = 3x^2 + 3x - 7$. Find the x -intercepts of the parabola in exact form.

Graphing quadratics

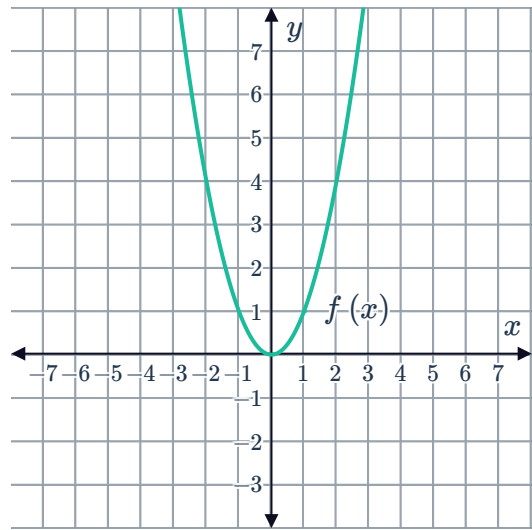
- 22 Consider the parabola described by the function $y = -2x^2 + 2$.

- a Is the parabola concave up or down?
- b Is the parabola more or less steep than the parabola $y = x^2$?
- c What are the coordinates of the vertex of the parabola?
- d Sketch the graph of $y = -2x^2 + 2$.

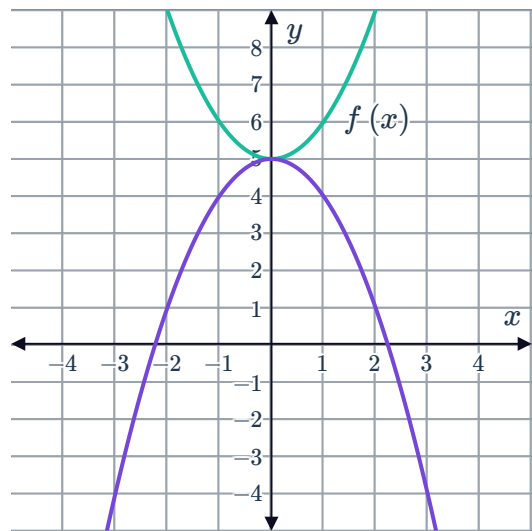


- 23** A graph of $f(x) = x^2$ is shown on the right. Sketch the graph after it has undergone transformations resulting in the following functions:

- a** $g(x) = 4x^2 - 2$
- b** $h(x) = 2(x + 4)^2$



- 24** Consider the two graphs. One of them has equation $f(x) = x^2 + 5$. What is the equation of the other graph?



- 25** Consider the quadratic function $h(x) = x^2 + 2$.

- a** Sketch the graph of the parabola $h(x)$.
- b** Plot the axis of symmetry of the parabola on the same graph.
- c** What is the vertex of the parabola?

- 26** Consider the parabola described by the function $y = \frac{1}{2}(x - 3)^2$.

- a** Is the parabola concave up or down?
- b** Is the parabola more or less steep than the parabola $y = x^2$?
- c** What are the coordinates of the vertex of the parabola?
- d** Sketch the graph of $y = \frac{1}{2}(x - 3)^2$.



27 Consider the equation $y = (x - 3)^2 - 1$.

- a Find the x -intercepts.
- b Find the y -intercept.
- c Determine the coordinates of the vertex.
- d Sketch the graph.

28 Consider the quadratic function $f(x) = -3(x + 2)^2 - 4$.

- a What are the coordinates of the vertex of this parabola?
- b What is the equation of the axis of symmetry of this parabola?
- c What is the y -coordinate of the graph of $f(x)$ at $x = -1$?
- d Sketch the graph of the parabola.
- e Plot the axis of symmetry of the parabola on the same graph.

29 On a number plane, sketch the shape of a parabola of the form $y = a(x - h)^2 + k$ that has the following signs for a, h and k :

- a $a > 0, h > 0, k > 0$
- b $a < 0, h > 0, k > 0$
- c $a > 0, h > 0, k < 0$
- d $a < 0, h > 0, k < 0$

30 Consider the parabola $y = (2 - x)(x + 4)$.

- a State the y -intercept.
- b State the x -intercepts.
- c Complete the table of values:
- d Determine the coordinates of the vertex of the parabola.
- e Sketch the graph of the parabola.

x	-5	-3	-1	1	3
y					

31 Consider the parabola $y = (x - 3)(x - 1)$.

- a Find the y -intercept.
- b Find the x -intercepts.
- c State the equation of the axis of symmetry.
- d Find the coordinates of the turning point.
- e Sketch the graph of the parabola.



 **32** Consider the parabola $y = x(x + 6)$.

- a Find the y -intercept.
- b Find the x -intercepts.
- c State the equation of the axis of symmetry.
- d Find the coordinates of the turning point.
- e Sketch the graph of the parabola.

 **33** Sketch the graph of the following:

a $y = (x + 2)(x - 3)$

b $y = (x - 3)(x + 1)$

 **34** Consider the function $y = (x + 5)(x + 1)$.

- a Sketch the graph.
- b Sketch the graph of $y = -(x + 5)(x + 1)$ on the same set of axes.

 **35** Consider the equation $y = x^2 - 6x + 8$.

- a Factorise the expression $x^2 - 6x + 8$.
- b Hence, or otherwise, find the x -intercepts of the quadratic function $y = x^2 - 6x + 8$
- c Find the coordinates of the turning point.
- d Sketch the graph of the function.

 **36** Consider the parabola $y = x^2 + x - 12$.

- a Factorise the equation of the parabola.
- b Find the x -intercepts of the curve.
- c Find the y -intercept of the curve.
- d What is the equation of the vertical axis of symmetry for the parabola?
- e Find the coordinates of the vertex of the parabola.
- f Is the parabola concave up or down?
- g Sketch the graph of $y = x^2 + x - 12$.



- 37** A parabola has the equation $y = x^2 + 4x - 1$.
- Express the equation of the parabola in the form $y = (x - h)^2 + k$ by completing the square.
 - Find the y -intercept of the parabola.
 - Find the vertex of the parabola.
 - Is the parabola concave up or down?
 - Hence, sketch the graph of $y = x^2 + 4x - 1$.
- 38** Consider the quadratic $y = x^2 - 12x + 32$.
- Find the zeros of the quadratic function.
 - Express the equation in the form $y = a(x - h)^2 + k$ by completing the square.
 - Find the coordinates of the vertex of the parabola.
 - Hence, sketch the graph.
- 39** Consider the curve $y = x^2 + 6x + 4$.
- Determine the axis of symmetry.
 - Hence, determine the minimum value of y .
 - Sketch the graph of the function.
- 40** Consider the function $P(x) = -2x^2 - 8x + 2$.
- Find the coordinates of the vertex.
 - Sketch the graph.
- 41** Consider the equation $y = 6x - x^2$.
- Find the x -intercepts of the quadratic function.
 - Find the coordinates of the turning point.
 - Sketch the graph.
- 42** A parabola is described by the function $y = 2x^2 + 9x + 9$.
- Find the x -intercepts of the parabola.
 - Find the y -intercept for this curve.
 - Find the axis of symmetry.
 - Find the y -coordinate of the vertex of the parabola.
 - Sketch the graph.



Graphing quadratics using technology

43 Use your calculator to graph the equations below. Then answer the following questions:

- i What is the vertex of the graph? ii What is the y -intercept?

a $y = 4x^2 - 64x + 263$

b $y = -4x^2 - 48x - 140$

44 Use your calculator to graph $y = -3x^2 - 12$.

- a What is the vertex of the graph?
b Are there any x -intercepts?
c For what values of x is the parabola decreasing?

45 Using a graphing calculator, sketch the curve $y = x^2 + 6.2x - 7$.

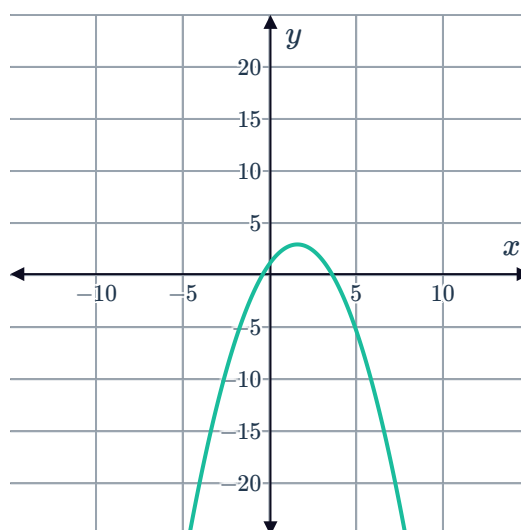
- a Determine the axis of symmetry. b Determine the minimum value of y .

46 Use a graphing calculator to sketch the parabola $y = -2x^2 + 16x - 24$.

- a Find the x -intercepts of the parabola.
b Find the y -intercept of the parabola.
c Find the axis of symmetry of the parabola.
d Find the y -coordinate of the vertex of the parabola.

47 Consider the function $y = -0.72x^2 + \sqrt{5}x + 1.21$.

- a Find the x -coordinate of the vertex to two decimal places.
b Find the y -coordinate of the vertex to two decimal places.
c Is the graph shown a possible viewing window on your calculator that shows the vertex and the x -intercepts?
d Use a graphing calculator to find the x -intercepts to two decimal places.



48 Consider the function $y = 0.91x^2 - 5x - 4$ 

- a Find the x -coordinate of the vertex to two decimal places.
- b Find the y -coordinate of the vertex to two decimal places.
- c Is the graph shown a possible viewing window on your calculator that shows the vertex and the x -intercepts?
- d Use a graphing calculator to find the x -intercepts to two decimal places.

